

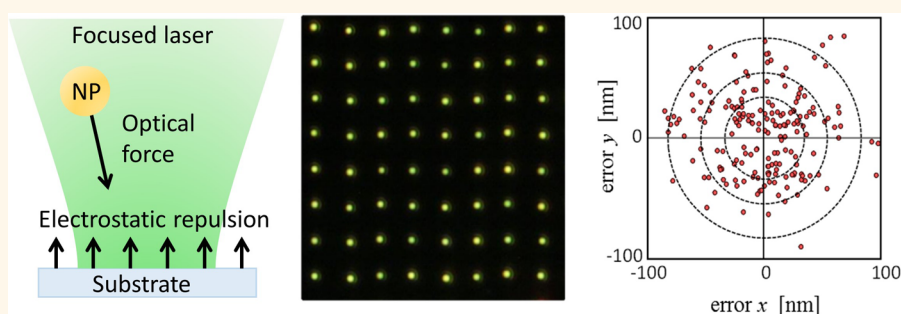
Accuracy and Mechanistic Details of Optical Printing of Single Au and Ag Nanoparticles

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ABSTRACT: Optical printing is a powerful all-optical method that allows the incorporation of colloidal nanoparticles (NPs) onto substrates with nanometric precision. Here, we present a systematic study of the accuracy of optical printing of Au and Ag NPs, using different laser powers and wavelengths. When using light of wavelength tuned to the localized surface plasmon resonance (LSPR) of the NPs, the accuracy improves as the laser power is reduced, whereas for wavelengths off the LSPR, the accuracy is independent of the laser power. Complementary studies of the printing times of the NPs reveal the roles of Brownian and deterministic motion. Calculated trajectories of the NPs, taking into account the interplay between optical forces, electrostatic forces, and Brownian motion, allowed us to rationalize the experimental results and gain a detailed insight into the mechanism of the printing process. A clear framework is laid out for future optimizations of optical printing and optical manipulation of NPs near substrates.

KEYWORDS: nanofabrication, optical forces, optical tweezers, colloidal patterning

An enormous and continuously growing list of colloidal nanoparticles (NPs) can be prepared by chemical methods. NPs made of metals, semiconductors, oxides, or magnetic materials can be synthesized with many different sizes, shapes, and compositions.^{1–7} This, in turn, gives rise to NPs with various physical properties such as optical, catalytic, electronic, and magnetic properties, which are size and shape dependent in the nanoscale.⁸ Also, the capping molecules of colloidal NPs can be exchanged or modified with a variety of molecular moieties, allowing the tailored design of their surface chemical or biochemical function.⁹ In summary, chemical synthesis permits the large-scale and low-cost production of colloidal NPs with a vast diversity of custom-made physical, chemical, and biological properties. However, colloidal NPs are obtained in liquid suspensions, and their assembly onto precise positions of a substrate is not simple.

Developing methodologies to control the incorporation of colloidal NPs onto substrates with nanometric precision is one

of the open challenges of nanotechnology. Such a method would enable the application on nano- and microdevices of the enormous assortment of physicochemical properties offered by colloidal NPs, which is unattainable by top-down methods.

To this end, numerous approaches have been developed.^{10,11} Convective self-assembly¹² or Langmuir–Blodgett¹³ methods allow the fast coverage of surfaces over macroscopic dimensions, but NPs organize according to their thermodynamically most stable configurations, and therefore their versatility to form arbitrary arrays is limited. Template-assisted methods such as capillary assembly¹⁴ or chemically patterned surface^{15,16} add flexibility of pattern design, but require additional steps that lengthen and complicate the fabrication

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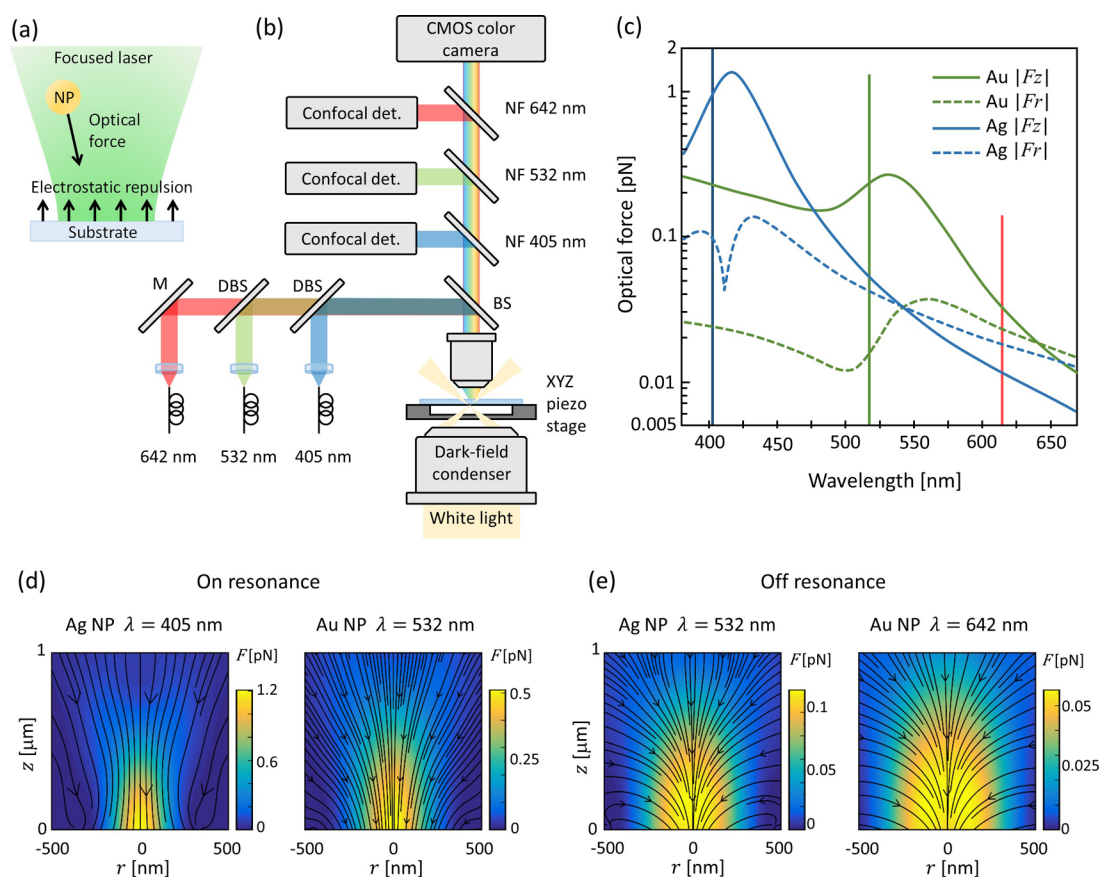


Figure 1. (a) Schematic of the optical printing process. (b) Scheme of the experimental setup. M: Mirror; BS: Beam splitter; DBS: Dichroic beam splitter; NF: Notch filter. (c) Absolute value of the maximum axial (F_z) and radial (F_r) forces versus wavelength for 60 nm Ag and Au NPs in a diffraction limited Gaussian focus with NA = 1.0 and an incident laser power of $P = 1$ mW. The three vertical lines correspond to the printing wavelengths used in the experiments: 405, 532, and 642 nm. (d, e) Force maps for 60 nm Au and Ag NPs near a Gaussian focus of different wavelengths. The arrows denote the direction of the force, and its magnitude is color coded.

procedure. Combination of multiple types of particles is also difficult with these methods.

Optical printing appears as a versatile alternative, where optical forces are used to directly capture colloidal NPs from suspension and immobilize them one by one on specific locations of a substrate.^{17–23} Although in principle it is applicable to any colloidal NP, optical printing is particularly suitable for metallic NPs due to their localized surface plasmons resonances (LSPR) which strongly enhance the interaction with light.²⁴ Well-defined arrays of metallic NPs have been fabricated by sequential optical printing of single spherical Au NPs,^{17,18} aligned Au nanorods,²¹ connected Au–Ag heterodimers,²⁵ gold bipyramids selected from suspension by the same printing light,²³ and gold NPs holding DNA-origami structures.²² Simultaneous optical printing of multiple NPs in an ordered fashion is also possible using spatially modulated light fields.^{20,26} Optical printing is also useful for the incorporation of NPs in predetermined positions of other structures such as photonic crystal cavities²⁷ and even biological cells.²⁸ Also, since the interaction of light with metallic NPs is size and composition dependent, optical printing with lasers of different wavelengths offers selectivity for the deposition of different kinds of NPs from a single suspension.²⁹

Despite of being an optical method, optical printing has demonstrated an accuracy of roughly 50–100 nm,^{17,18} well beyond the diffraction limit. However, the mechanism and accuracy of optical printing have been so far discussed only

superficially.^{17,18} Briefly, optical printing is based at least on two phenomena: colloidal interaction between the NPs and the substrate and optical forces acting on the NPs¹⁷ (Figure 1a). A substrate with the same sign of charge as the NPs is used. A DLVO potential between NPs and substrate prevents spontaneous binding. Optical forces are applied to overcome the repulsion barrier on specific locations of the substrate and immobilize the NPs there by van der Waals interaction. Still, several key questions remain open such as what is the best attainable precision to immobilize a NP? What are the limiting factors for that accuracy? The role of the LSPRs is also unclear on this respect. In different experiments, Au and Ag NPs have been immobilized by optical printing using light of wavelength tuned to the LSPR^{17,25,26,30} or off-resonance,^{20,27} which corresponds to qualitatively different force fields.

Here, we present a systematic investigation of the accuracy of optical printing to immobilize a single NP in a target position, as a function of various experimental parameters, such as the laser power and wavelength, and the NP material. Au and Ag NPs of 60 nm in diameter were optically printed using different irradiance levels tuned at and off their LSPR. A comparison of the experimental observations to calculations provides a detailed insight into the mechanism of optical printing and the parameters influencing the printing accuracy.

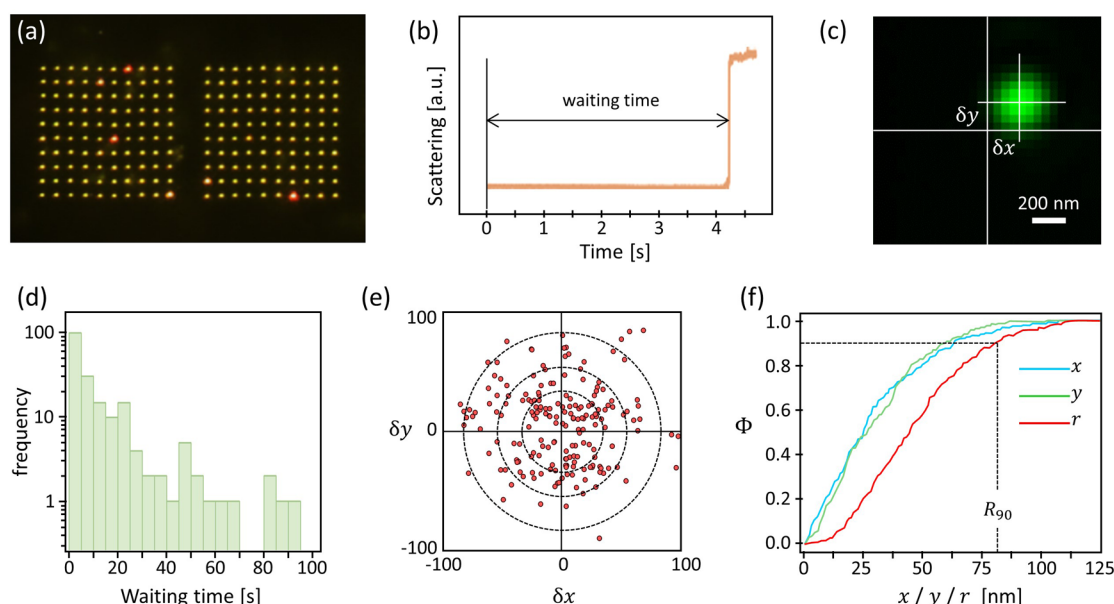


Figure 2. (a) Dark-field image of two square (10×10) arrays of optically printed 60 nm Au NPs. (b) Time-trace of the confocal scattering signal during optical printing of a NP. Upon printing of a NP, the signal shows a stepwise increase. (c) Example of a scattering confocal image used to determine the printing error δx and δy . (d) Histogram of waiting times for the optical printing of 196 Au NPs ($P = 1.17$ mW, $\lambda = 532$ nm). The median is 4.7 s. (e) 2D scatter plot of the printing error of the 196 printed Au NPs. The circles delimit the regions of the xy plane, where 33%, 66%, and 90% of the NPs were printed. (f) 1D (x and y) and radial (r) cumulative distribution functions.

RESULTS AND DISCUSSION

Multicolor Optical Printing Setup. Optical printing was carried out on a modified upright dark-field microscope as schematically shown in Figure 1b. CW lasers with wavelengths of 405, 532, and 642 nm were used for optical printing and confocal detection of elastically scattered light. The latter enabled the precise determination of the positions of the printed NPs. In all cases, the lasers were focused nearly to their diffraction limit and are defined by their wavelength λ , their Gaussian beam waist w_0 , and total power P . All relevant functions of the microscope such as stage positioning and control of the lasers were automated. Further details on the instrumentation can be found in our previous work.²⁵

On and off Resonance Optical Forces on Metallic NPs. During optical printing, the action of optical forces can be divided into radial forces that push the NP toward the center of the beam and vertical forces that push it toward the substrate. The strength and direction of the optical forces depend on parameters of both the NP and the optical field. The size, shape, and material composition define the NP polarizability, which in combination with the intensity gradients and wavelength of the optical field determine the optical force.

To illustrate this, Figure 1c shows, as a function of wavelength, the modulus of the maximum axial and radial forces exerted on Ag and Au NPs of the same diameter (60 nm), placed on a diffraction limited focus ($NA = 1.0$). The magnitude of optical forces is many times stronger for wavelengths close to the plasmon resonance of the NPs. Also, the ratio between the radial and the axial forces changes dramatically with the wavelength. On resonance, axial forces dominate, being approximately 10-fold stronger than radial forces. On the contrary, for wavelengths out of the LSPR toward the red region, both axial and radial forces have comparable magnitudes. Figures 1d and 1e show representative force maps for Ag and Au at laser wavelengths on and off

resonance, respectively. Details on force calculations are provided in the Materials and Methods section.

Automated Characterization of Optical Printing. Two parameters were used to characterize the process of optical printing: the waiting time until a NP is printed, and the printing accuracy. For each condition of optical printing the distributions of waiting times and of printing error were obtained from the analysis of a large number of printing events according to the following procedure. For a given type of NP, laser power, and wavelength, 200 NPs were optically printed in two square grids of 10×10 NPs. Figure 2a shows an example of such grids of Au NPs. Eventually, faulty printing events occur, such as the four reddish NPs on the grids of Figure 2a. Such NPs, which are either nonspherical or larger in this case, were excluded from the analysis.

For each one of the printed NPs, the waiting time and printing error were measured as follows. First, the stage was moved to the target position for the optical printing of a NP (x_0, y_0). The selected laser beam was unblocked, and the time-trace of the corresponding confocal scattering signal was monitored and analyzed online. Printing events were detected as a stepwise increase in the scattering signal, as shown in Figure 2b. The waiting time is defined as the time interval of illumination until a NP is printed. Right after the printing of a NP, a confocal scattering image is acquired using a sufficiently lower laser intensity so as to prevent further printing events during scanning. The scanning range was centered at the target position (x_0, y_0). The center of mass of the image was computed and assigned to the position of the printed NP (x_i, y_i). The difference $(\delta x_i, \delta y_i) = (x_i - x_0, y_i - y_0)$ is defined as the printing error along x and y axis for i^{th} event (Figure 2c). Using this approach, errors due to mechanical drift of the microscope, typically below 10 nm/min, are negligible because acquiring an image takes <10 s.

An example of the waiting time distribution is shown in Figure 2d. Typically the waiting times do not follow a purely

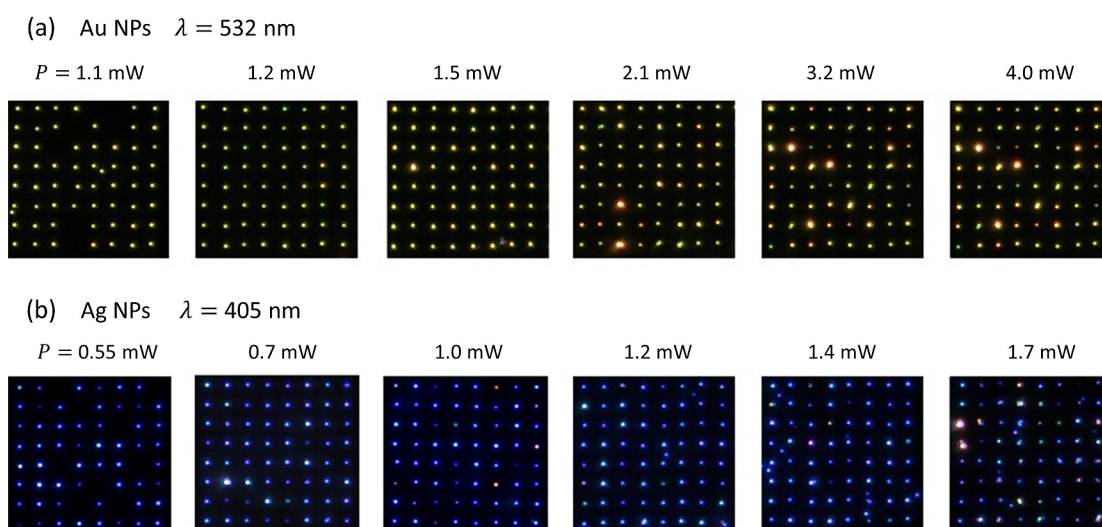


Figure 3. On-resonance printing: Example arrays of (a) Au and (b) Ag 60 nm NPs fabricated through on-resonance optical printing at different laser powers.

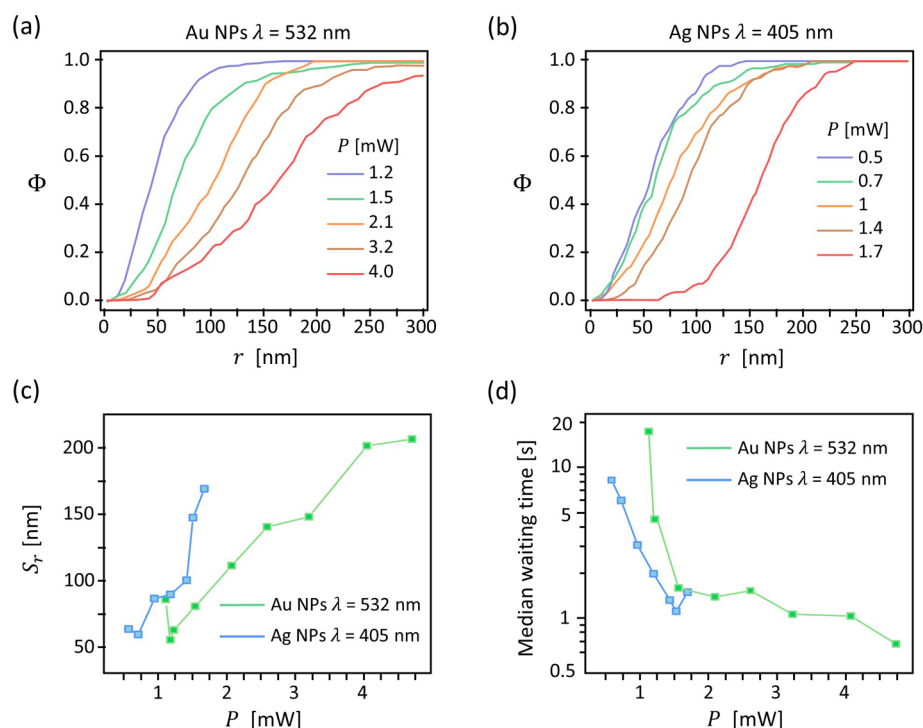


Figure 4. On-resonance optical printing. (a, b) Radial cumulative probability distribution $\Phi(r)$ for Au (a) and Ag (b) NPs printed with different powers with wavelengths at their respective LSPR. (c, d) Radial root-mean-square deviation S_r (c) and median waiting time (d) as a function of laser power.

exponential distribution as expected for a totally stochastic and memoryless process. Therefore, the median waiting time was used as a characteristic of each condition for optical printing.

The optical printing accuracy can be visualized by means of 2D scatter plots of the printing error, as shown in the example of Figure 2e. The concentric circles represent the areas containing 33%, 66%, and 90% of the NPs. The radius containing 90% of the NPs is called R_{90} . An alternative way of visualizing the same information is through the cumulative probability distribution function Φ , that is, the probability that a NP will be printed within a certain range from the target coordinate. Figure 2f shows the 1D cumulative probability

distribution functions $\Phi(x)$ and $\Phi(y)$. Both functions are practically identical, indicating that the printing error is isotropic. Furthermore, since the NPs were printed with linearly polarized light along the x axis, printing error is independent of the polarization direction. The standard deviations of these distributions are $S_x = 38.8$ nm and $S_y = 37.6$ nm. In the following, the printing accuracy will be discussed in terms of the radial cumulative probability distribution $\Phi(r)$ (Figure 2f) and the radial root-mean-square deviation, which in the present example is $S_r = 53.9$ nm.

On-Resonance Optical Printing. The influence of laser power on the accuracy of optical printing near LSPR was

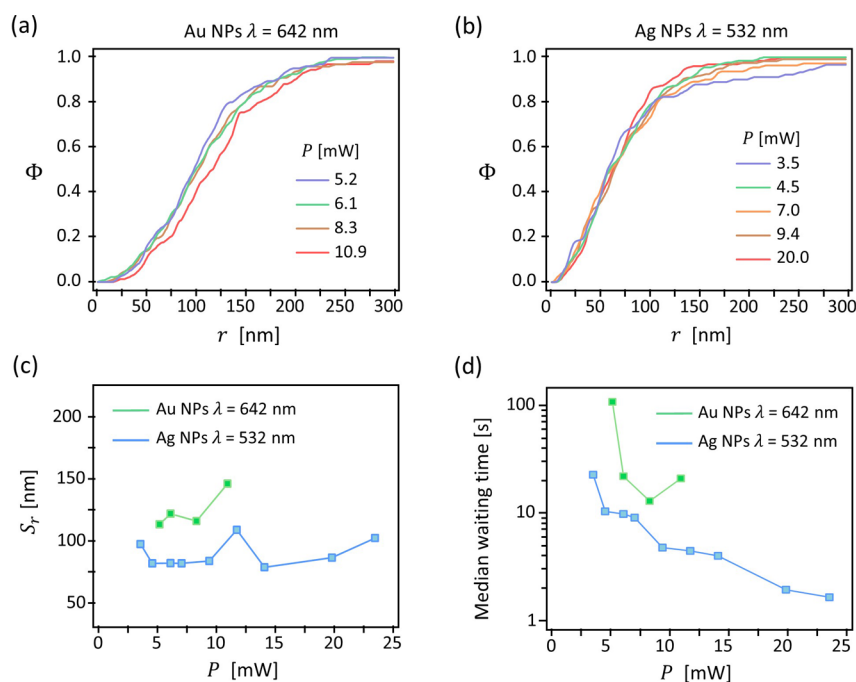


Figure 5. Off-resonance optical printing. (a, b) Radial cumulative probability distribution $\Phi(r)$ for Au (a) and Ag (b) NPs printed at 642 and 532 nm, respectively. (c, d) Radial root-mean-square deviation S_r (c) and median waiting time (d) as a function of laser power.

investigated for the two types of NPs. The 60 nm Au NPs were printed with a laser focus of wavelength $\lambda = 532$ nm and beam waist $w_0 = (265 \pm 5)$ nm. The 60 nm Ag NPs were printed with a laser focus of $\lambda = 405$ nm and $w_0 = (226 \pm 7)$ nm. Capping layer of the Au and Ag NPs is citrate in both cases, and the same surface functionalization was carried out on the substrate before printing. Example grids of printed NPs are shown in Figure 3a,b. The results were similar for both materials. For both NPs, there is a limited range of laser power that is useful for optical printing. At lower powers, printing events become increasingly infrequent; the optical forces become too weak to overcome the electrostatic repulsion. A maximum waiting time of 2 min before moving to the next position was set, and this is the reason why some target positions of the grids remain empty at the lowest powers. If the laser power is decreased below a certain threshold, no particle is printed within any practical period of time. A threshold power P_{th} was defined as the minimum power for which there is at least one printing event in 10 min. It was found to be $P_{th} = (0.55 \pm 0.05)$ mW for the Ag NPs and $P_{th} = (1.00 \pm 0.05)$ mW for the Au NPs.

As the laser power increases, so does the number of defective NPs. Those are the NPs appearing with different colors in the dark-field images of the example grids (Figure 3). This could be caused by the loss of selectivity as the optical forces became strong enough to print other types of NPs from the suspension. Another possibility is that the NPs undergo a change of shape induced by the high temperatures and/or electric fields generated under illumination at the LSPR.^{31,32} The power was increased until the defective NPs outnumbered the regular NPs.

Figure 4a,b shows $\Phi(r)$ obtained for Au and Ag NPs printed with five different laser powers. For both NPs, the precision of optical printing becomes poorer as the laser power increases. Figure 4c summarizes this observation displaying the radial root-mean-square deviation as a function of laser power for both materials. The precision improves monotonically as the

laser power is decreased, reaching values of around 50 nm, which is 4–5 times smaller than the beam waist of the laser foci used. However, using laser powers very close to the threshold of optical printing leads to slightly larger printing errors.

The waiting times show a strong dependency on laser power. Figure 4d shows the median waiting times for the optical printing of Ag and Au NPs as a function of laser power. In both cases the waiting time increases as the laser power reduces, becoming drastically longer as the threshold approaches.

Off-Resonance Optical Printing. The 60 nm Au NPs were printed with a focused laser of $\lambda = 642$ nm and $w_0 = (319 \pm 5)$ nm. The 60 nm Ag NPs were printed with $\lambda = 532$ nm and $w_0 = (265 \pm 5)$ nm. As expected from the smaller polarizabilities involved, higher laser powers are required for optical printing. The threshold powers were in this case $P_{th} = (5.0 \pm 0.5)$ mW for Au and $P_{th} = (3.5 \pm 0.5)$ mW for Ag.

The automation of the off-resonance measurements required a few adjustments. On resonance, the automatic detection of printing events is done using the scattering signal of the printing beam. When working off-resonance, the scattering signal of the NPs is weak, and the signal on the photodiodes may be very small or negative due to interference with the field reflected at the water–glass interface.³³ For this reason, a detection laser tuned to the LSPR of the NPs was superimposed to the printing beam. The intensity of the detection laser was kept as low as possible, and in all cases the force exerted by this beam was below 10% of the force exerted by the printing beam. Also, due to the strong radial forces present when using off-resonance light (Figure 1e), NPs were in some cases momentarily trapped in the confocal volume, raising the detection signal. For this reason an additional confirmation step was included. After a NP was detected, both the printing and the detection beams were blocked for a time interval of 1 s. Then, the presence of the NP was interrogated with the detection beam alone, in order to check whether the particle was fixed on the surface. If the NP was not fixed, then

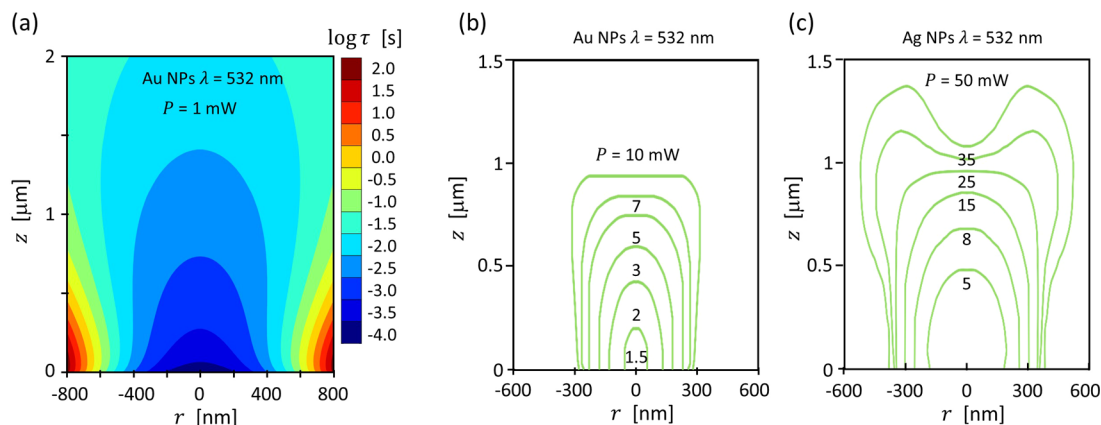


Figure 6. (a) Time-of-flight for deterministic motion for a 60 nm Au NPs with a laser beam of $\lambda = 532$ nm, $w_0 = 265$ nm, and $P = 1$ mW. (b, c) Capture surfaces S_C of $Pe = 1$ for different powers levels, calculated at $\lambda = 532$ nm for printing Au NPs on-resonance (b) and Ag NPs off-resonance (c). A beam waist of $\lambda = 265$ nm was used.

the printing beam was unblocked, and the process was repeated until a NP was effectively printed. Once the printing was confirmed, the waiting time was stored, and the position of the printed NP and the printing error were determined.

Figure 5a,b shows $\Phi(r)$ obtained for Au and Ag NPs printed with different powers of light off-resonance. Remarkably, the precision of optical printing off-resonance is practically independent of the laser power used. Although the precision in this case is slightly poorer than the best precision attained in resonance, it remains below the diffraction limit. The average S_r for the Au NPs was (124 ± 15) nm and for the Ag NPs (91 ± 10) nm (Figure 5c). These values are 2.6 and 2.9 times smaller than the corresponding beam waist used in each case. The waiting times are shown in Figure 5d. They are in general larger than the ones observed on resonance, but follow the same decreasing tendency with increasing power.

Mechanism of Optical Printing. *Waiting Time for Optical Printing.* During the optical printing process, the NPs may experience two kinds of motion: (i) Deterministic motion driven by the external optical field or (ii) diffusive or Brownian motion.

Deterministic motion in fluid flows can be classified according to the Reynolds number:³⁴

$$Re = \frac{\rho \tilde{v} \tilde{L}}{\mu} \quad (1)$$

where $\tilde{v}$ is a characteristic velocity, $\tilde{L}$ a characteristic length, μ the dynamic viscosity, and ρ the fluid density. The Reynolds number estimates the ratio between inertial and viscous forces. If $Re \ll 1$, viscous forces dominate. Considering water flow around one of the Au or Ag NPs, using $\tilde{L} = 60$ nm, $\mu = 10^{-3}$ Pa s, and $\rho = 10^{-3}$ kg/m³, a value of $Re = \tilde{v} 6 \times 10^{-2}$ s/m is obtained, which is $\ll 1$ for any physically meaningful velocity of the NPs during the optical printing process. The movement of a NP in a flow with such a small Re is overdamped. Then, during the optical printing, the NPs moves at the terminal velocity corresponding to the applied optical force, according to Stokes law. A NP near the focus will experience a position-dependent optical force $F_{op}(r)$, which determines a position-dependent velocity $v(r)$:

$$v(r) = \frac{F_{op}(r)}{6\pi\mu a} \quad (2)$$

where a is the NP radius and $\mathbf{r} = (x, y, z)$ is the tridimensional position vector. It is then possible to assign a characteristic time for the deterministic motion of a NP at each position:

$$\tilde{\tau}_d(r) = \frac{\tilde{L}}{|v(r)|} = \frac{\tilde{L}6\pi\mu a}{F_{op}(r)} \quad (3)$$

Under these conditions the optical force field lines can be interpreted as velocity lines or trajectories. Given an initial position $\mathbf{r}_0$, the trajectory toward the surface is determined by the optical forces, and a “time of flight” can be calculated by the path integral on the trajectory from that position $\mathbf{r}$ to the substrate:

$$\tau(r) = \int_r^{z=0} \frac{6\pi\mu a}{F_{op}(r)} dl \quad (4)$$

Figure 6a shows the time-of-flight for 60 nm Au NPs particles near the Gaussian focus of $\lambda = 532$ nm, $w_0 = 265$ nm, and $P = 1$ mW, which are typical parameters of optical printing on resonance. The predicted motion of the Au NPs under these conditions is much faster than the observed waiting times. For example, a particle starting 2 μ m away from the surface at the center of the beam has a time-of-flight of 10 ms, while the observed waiting times are typically seconds. Therefore, the optical printing process cannot be explained only in terms of deterministic motion, and Brownian motion must be involved as well.

The mean square displacement of a freely diffusing particle in three dimensions in a time t is given by

$$\langle \Delta r^2 \rangle = 6Dt = \frac{k_B T}{\pi\mu a} t \quad (5)$$

where $D = k_B T / (6\pi\mu a)$ is the diffusion coefficient, k_B is the Boltzmann constant, and T is the temperature. The average time that a NP needs to cover some characteristic distance $\tilde{L}$ by free diffusion is

$$\tilde{\tau}_f(r) = \frac{\tilde{L}^2 \pi\mu a}{k_B T(r)} \quad (6)$$

Considering a NP with diameter of 60 nm, in water and at 25 °C, the time needed by the NP to diffuse over a length equal to its own radius is $\tilde{\tau}_f = 6 \times 10^{-5}$ s. However, during the printing process, the characteristic time for Brownian motion $\tilde{\tau}_f$ will vary

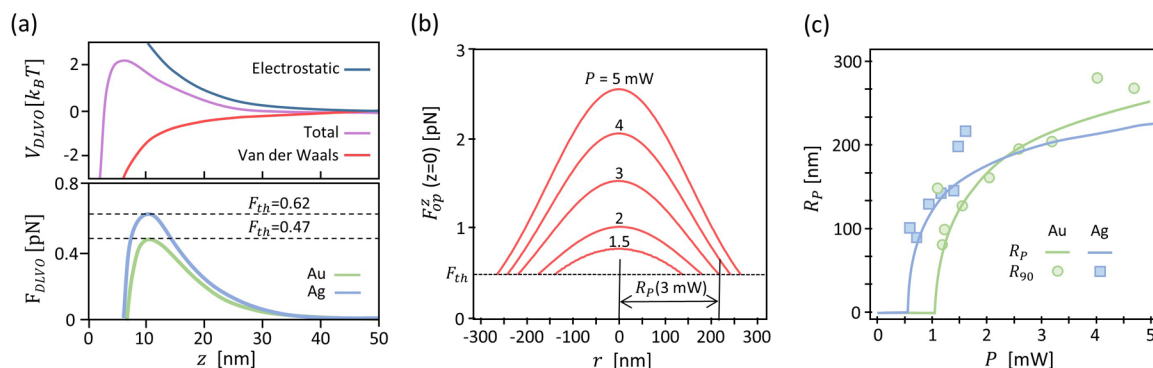


Figure 7. (a) Calculated DLVO potentials (top, 60 nm Au NP) and force (bottom, 60 nm Au and Ag NPs) of a NP versus the distance to a flat charged surface. Maximum repulsive forces (DLVO threshold barriers) are indicated with horizontal dashed lines. (b) Calculated optical force F_{op}^z at the substrate (focal plane, $z = 0$) as a function of the radial position, for Au NPs at $\lambda = 532$ nm and different beam powers. The horizontal line indicates the threshold for optical printing imposed by the repulsive DLVO barrier F_{th} . The region $F_{op}^z > F_{th}$ is delimited by R_p , schematically shown for $P = 3$ mW. (c) Solid lines: R_p as a function of laser power P for 60 nm Au and Ag NPs for the conditions of optical printing on-resonance. Markers: experimental values of R_{90} for Au (circles) and Ag (squares) NPs printed on-resonance.

due to position-dependent light absorption and heating of the NP. We accounted for this effect incorporating the surface temperature of the NP at each position within the laser focus. More accurate predictions could be done considering the temperature profile around the NP and the temperature dependence of the water viscosity.³⁵

If the characteristic time for free diffusion is much shorter than its deterministic counterpart, the NP motion will be purely Brownian. If the opposite happens, the motion of the NP will be deterministic dominated by the force field. One way to analyze this situation is by means of the dimensionless Peclet number:

$$Pe = \frac{\tilde{\tau}_f}{\tilde{\tau}_d(\mathbf{r})} = \frac{\tilde{L}|F_{op}(\mathbf{r})|}{6k_B T(\mathbf{r})} \quad (7)$$

If the Pe is much lower (higher) than unity, diffusion is dominant (negligible). Using Pe , it is possible to define a “capture volume” V_C , which is the region of space where the optical forces are stronger than Brownian motion. This volume and its boundary surface S_C are defined as

$$V_C = \{\mathbf{r}: Pe(\mathbf{r}) > 1\} \quad (8)$$

$$S_C = \{\mathbf{r}: Pe(\mathbf{r}) = 1\} \quad (9)$$

Figure 6b,c shows the capture surfaces S_C for Au NPs on-resonance and for Ag NPs off-resonance for different laser powers. In both cases a focused beam of $\lambda = 532$ nm with a waist of $w_0 = 265$ nm was used. The capture volume increases with laser power. If a particle is inside this volume, then it is driven toward the substrate by the optical force field.

The waiting times correspond to the time needed by the first NP to reach S_C by diffusive motion. Since it depends on the initial position of the particle and its random motion, the waiting time is a stochastic variable with values varying from one event to another. The fact that the capture volume increases with the power explains why the waiting time decreases. It is important to note that the concentration of the colloids used is approximately 2×10^9 NPs/mL. The volume per NP is $500 \mu\text{m}^3$, which is greatly larger than V_C . For example for the Au NPs, $\lambda = 532$ nm and $P = 1.5$ mW, $V_C = 0.05 \mu\text{m}^3$. That means that V_C is most probably empty of NPs when the printing laser is switched on.

Printing Error. The width of the capture volume is comparable to the beam waist w_0 , that is, a few hundreds of nanometers (Figure 6b,c). However, NPs can be printed in a predetermined position with root-mean-square deviations as small as 50 nm. In order to explain this, it is important to recall that there is an electrostatic repulsion barrier between the NPs and the substrate at very short separations. This repulsion can be modeled with the DLVO theory (details about these calculations are provided in the Materials and Methods section). Figure 7a shows the DLVO potentials for a 60 nm Au NP, along with the relevant forces for 60 nm Au and Ag NPs, both as a function of separation distance to the substrate. According to this model, the electrostatic barrier is around $2k_B T$, and the maximum repulsive force is $F_{th} = 0.47$ pN for Au NPs, and $F_{th} = 0.62$ pN for Ag NPs. These repulsion forces define a minimum threshold optical force for printing. The necessary condition for printing a NP in a position r on the substrate demands the following condition:

$$F_{op}^z(r, z = 0) > -F_{th} \quad (10)$$

Considering that the range of action of the repulsion is limited to a few nanometers from the surface and that these distances are short compared to the typical variation lengths of the optical fields, F_{op} was evaluated at the surface, that is, the focal plane $z = 0$. Condition 10 (eq 10) can be assessed theoretically by contrasting calculated optical and interaction forces between the NPs and the substrate. For each laser power, a “printing radius” R_p is defined as the radius of the region in the focal plane where the axial optical force is above the threshold, that is, the radius within which the axial optical force fulfills condition 10 (eq 10). Figure 7b shows F_{th} and the axial optical force $F_{op}^z(r, z = 0)$ for different laser powers calculated for a 60 nm Au NPs in the focus of $\lambda = 532$ nm. The value for R_p for $P = 3$ mW is depicted. Figure 7c shows, as a function of the laser power, the calculated R_p for conditions of optical printing on-resonance of the 60 nm Au and Ag NPs, together with the experimental values of R_{90} . The agreement between the calculated R_p and experimental R_{90} as a function of laser power, including the threshold power, is remarkable, considering that there were no free parameters in the calculations. This indicates that optical printing on-resonance is well described by the simple balance of the axial optical force and the electrostatic repulsion threshold. There is a systematic

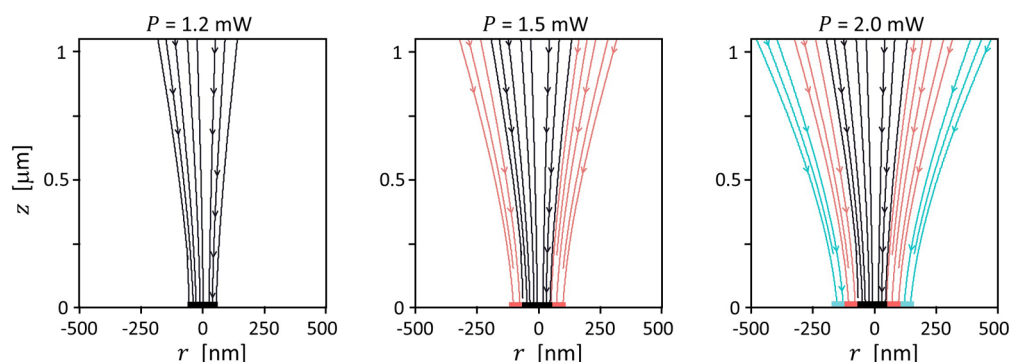
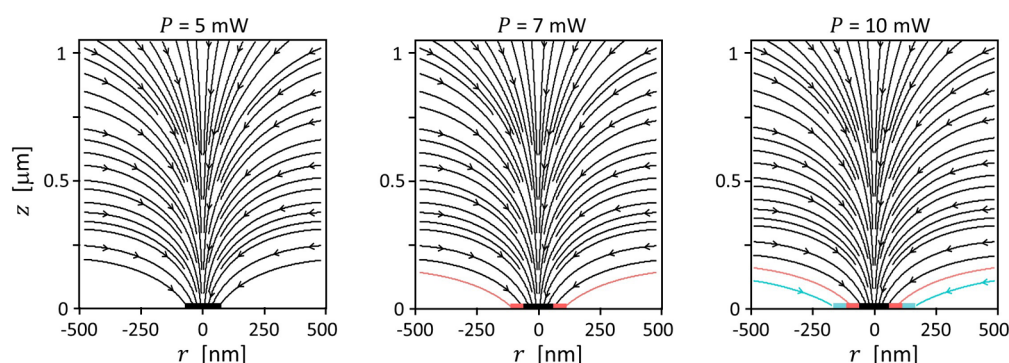
(a) Au NPs $\lambda = 532$ nm(d) Au NPs $\lambda = 532$ nm

Figure 8. Calculated trajectories for 60 nm Au NPs at different powers and wavelengths. Only the trajectories that end inside the printing radius R_p are shown. Pink and light blue lines are those trajectories that become available when increasing the power with respect to the lower value. Colored boxes at $z = 0$ depict the final position of the trajectories. The added size of the boxes correspond to R_p .

deviation at the highest laser powers, where the observed R_{90} is larger than R_p . This can be due to the temperature increase that the NPs experience at those high powers, which can induce a reduction of the repulsion.

At this point, it becomes clear why reducing the laser power is not a viable strategy to further increase the printing accuracy because close to P_{th} the value of R_p changes too fast for realistic changes of beam power (see Figure 7c). It must be noted that this limitation is not a matter of laser power stability. The colloidal NPs have finite size distribution, which translates in a distribution of optical forces. For example, Au NPs have a diameter distribution of $(63 \text{ nm} \pm 3 \text{ nm})$, which translate in variation of the maximum optical forces at 532 nm of more than 15%. In addition, the charge distribution of the substrate may present slight variations. Therefore, even for a perfectly stable laser source, the actual axial force exerted on the NP as well as the repulsion threshold will vary from one printing event to another. In practice, using laser powers very close to P_{th} leads to a slightly worse accuracy (compare Figures 4c and 7c). At these laser powers, printing events become less frequent, and it is possible that they become sensible to thermal fluctuations or end up biased toward regions of the substrate with local minima of charge density.

In the case of optical printing with light off-resonance, R_p is no longer a good representation of R_{90} , indicating that a deeper analysis is necessary. A more detailed discussion is possible by considering the deterministic trajectories of the NPs once they enter the capture volume. The end position of the trajectories on the substrate is determined by the force field and the

position where the NPs enter the capture volume. If the end position lies inside the printing radius R_p , the electrostatic barrier will be overcome, and the NP will be printed.

Figure 8 shows the calculated trajectories for 60 nm Au NPs when optically printed using different powers at $\lambda = 532$ nm (on-resonance) and $\lambda = 642$ nm (off-resonance). Only the trajectories that lead to optical printing are shown. For the case on-resonance (Figure 8a), radial optical forces are weak, and therefore the trajectories are mainly axial. Only NPs entering the capture volume from the top central region are able to reach the substrate. For a laser power just above threshold (1.2 mW), not many trajectories lead to printing. If the laser power is increased, more trajectories become available for printing, increasing the probability of printing a NP and the size of R_p , which explain the experimentally observed reduction of printing waiting time and accuracy.

The situation changes qualitatively if light off-resonance is used (Figure 8b). Completely different NP trajectories are involved because radial and axial forces are comparable in this case. Therefore, for a laser power just above threshold, radial forces are strong enough to drive NPs toward the substrate from a larger portion of the 2π solid angle above the substrate. As in the case on-resonance, if the power is increased, new printing trajectories become available. However, those additional trajectories cover a very small region of space, and the probability that NPs will diffuse into such an initial position is relatively small. Therefore, even though R_p is increased, only minor or negligible changes are expected on the experimental distribution of printing positions. This explains the independ-

ence of R_{90} on laser power observed off-resonance in Figure S4b.

CONCLUSIONS

Optical printing of metallic NPs was investigated experimentally as a function of laser power and wavelength for Au and Ag NPs.

Two different regimes were identified depending on whether the laser used is tuned to the LSPR of the NPs (on-resonance) or not (off-resonance). On-resonance, optical printing provides higher precision using lower laser powers, up to a limit where printing is not possible because the optical forces are not strong enough to overcome the electrostatic repulsion barrier by the substrate. Off-resonance, the precision is slightly poorer than the best precision attained on-resonance, but is practically independent of the laser power.

A detailed physical description of the process, including the calculation of optical forces, Brownian motion, and surface repulsion was contrasted to the experimental results of the precision and waiting times.

We introduced the concept of a “capture volume” where deterministic motion driven by optical forces overcomes Brownian motion. NPs perform diffusive motion until they reach this volume, where they are dragged toward the substrate by optical forces. Since the capture volume increases at higher powers, the waiting times are reduced. A rigorous balance between optical and DLVO interactions together with a detailed consideration of the amount and shape of deterministic trajectories inside the capture volume allowed us to quantitatively explain the measured precisions.

These investigations expand the understanding of the processes behind optical printing and constitute a useful guideline for the optimization of future experiments. Until now, implementations of optical printing have mainly used Ag and Au NPs. But the library of available colloidal NPs is huge, and the ability to predict optical forces on complex NPs is growing accordingly.³⁶ Considering the mechanistic details presented here, procedures for highly precise incorporation of a vast variety of NPs onto substrates can be designed, broadening the spectra of nanostructures that could be fabricated by means of optical forces.

MATERIALS AND METHODS

Sample Preparation. Au and Ag NPs were optically printed on glass substrates. Both NPs were citrate coated and negatively charged at neutral pH. Substrates were surface functionalized with negative charges using a layer-by-layer deposition of polyelectrolytes.³⁷ Polydiallyldimethylammonium chloride (PDPA) $M_w = 400,000$ –500,000 and sodium polystyrenesulfonate (PSS) $M_w = 70,000$ (both from Sigma-Aldrich), were used as polycation and polyanion, respectively. Glass coverslips were first cleaned through the following steps: sonication for 10 min in Hellmanex (0.2% v/v), thorough rinsing with Milli-Q water, rinsing with acetone, drying at 80 °C, and plasma cleaning for 3 min. Then, the substrate was surface modified by immersion for 15 min in PDPA solution (1 mg/mL in 0.5 M NaCl), rinsed with Milli-Q water, immersion for 15 min in PSS solution (1 mg/mL in 0.5 M NaCl), rinsed, and stored in Milli-Q water.

Citrate-capped Au NPs were purchased from Nanopartz (specified diameter: 62 nm $\pm$ 2 nm; average diameter determined by FE-SEM: 63 nm $\pm$ 3 nm). Citrate-capped Ag NPs were purchased from Sigma-Aldrich (specified diameter: 60 nm $\pm$ 8 nm; average diameter determined by FE-SEM: 57 nm $\pm$ 5 nm). Original colloids were diluted in NaCl 1.5 mM, to a final concentration of 2×10^9 NPs/mL.

Force Calculations. We performed two different calculations of the optical force fields. To evaluate general trends (Figures 1c), we

calculated optical forces on metallic NPs in the focus of a Gaussian beam under the paraxial and Rayleigh approximations according to Agayan *et al.*:³⁸

$$F_{\text{grad}}^z(r, z) = -\alpha' |E_0|^2 \frac{\epsilon_0 w_0^4}{\pi z_0^2} z \left(\frac{1}{w^4} - \frac{2r^2}{w^6} \right) e^{-2r^2/w^2}$$

$$F_{\text{grad}}^r(r, z) = -\alpha' |E_0|^2 \frac{2\epsilon_0 w_0^2}{\pi w^4} r e^{-2r^2/w^2}$$

$$F_{\text{scat}}^z(r, z) = \alpha'' |E_0|^2 \frac{\epsilon_0 w_0^2}{\pi w^2} \left(k - \frac{kr^2(z^2 - z_0^2)}{2(z^2 + z_0^2)^2} - \frac{w_0^2}{z_0 w^2} \right) e^{-2r^2/w^2}$$

$$F_{\text{scat}}^r(r, z) = \alpha'' |E_0|^2 \frac{\epsilon_0 k w_0^2}{\pi R w^2} r e^{-2r^2/w^2}$$

$$k = 2\pi n/\lambda \quad w(z) = w_0 \sqrt{1 + \frac{z^2}{z_0^2}} \quad R(z) = z \left(1 + \frac{z_0^2}{z^2} \right)$$

where w_0 is the beam waist at the focal plane $z = 0$, also called the Gaussian beam radius, $z_0 = \pi w_0^2 n/\lambda$, E_0 is the field amplitude at $(0,0)$, related to the total beam power through $|E_0|^2 = 2P/(n c \epsilon_0 w_0^2)$, ϵ_0 is the permittivity of vacuum, c corresponds to the speed of light in vacuum, and α' and α'' denote the real and imaginary part of particle polarizability, respectively. It was calculated using Kuwata expression:³⁹

$$\alpha = V \frac{1 - \frac{1}{10}(\epsilon + \epsilon_m)x^2}{\left(\frac{1}{3} + \frac{\epsilon_m}{\epsilon - \epsilon_m}\right) - \frac{1}{30}(\epsilon + 10\epsilon_m)x^2 - i \frac{4\pi^2 \epsilon_m^{3/2} V}{3 \lambda^3}}$$

where V is the volume of the sphere, $x = \frac{2\pi a}{\lambda_m}$ is the size parameter, and ϵ and ϵ_m are the dielectric functions of the particle and the medium, respectively.

For quantitative comparison to the experiments (Figures 1d,e, 6, and 7), we employed generalized Lorenz–Mie theory,⁴⁰ and we assumed spherical NP. The incident Gaussian beam was focused by an objective and formed a nonparaxial beam focus.⁴¹ From the best overlap between the measured and calculated focused beam radius we determined the filling factor (ratio of incident beam waist radius and aperture radius of the lens, (eq 3.55 in ref 41) and calculated the optical forces according to ref 42.

DLVO Calculations. The interaction between the NP and the substrate was calculated using the DLVO theory for a sphere and a plane surface.^{43,44} The total potential has two contributions, an electrostatic repulsion V_E and a van der Waals attraction V_{vdW} :

$$V_{\text{DLVO}} = V_E + V_{\text{vdW}}$$

The electrostatic potential is

$$V_E(z) = 16e^{-\kappa d} \epsilon_m a \left(\frac{\zeta e \psi_p}{\zeta e} \right)^2 \tanh \left(\frac{\zeta e \psi_p}{k_B T} \right) \tanh \left(\frac{\zeta e \psi_s}{k_B T} \right)$$

where κ is the inversed Debye length, e is the electron charge, ζ is the ion valency, ψ_p and ψ_s are the zeta-potential of the sphere and the surface, respectively.

The nonretarded van der Waals potential is

$$V_{\text{vdW}}(z) = -\frac{H}{6} \left(\frac{z}{a} + \frac{a}{2a+z} + \ln \frac{z}{2a+z} \right)$$

where H is the Hamaker constant, which for a dielectric and a metal interacting through a liquid can be expressed as⁴⁴

$$H = \frac{3}{8\sqrt{2}} \left(\frac{n_1^2 - n_3^2}{n_1^2 + n_3^2} \right) \frac{h\sqrt{\nu_1 \nu_3} \nu_m}{(\sqrt{\nu_1 \nu_3} + \nu_2/\sqrt{n_1^2 - n_3^2})}$$

where n_1 , n_3 , ν_1 , ν_3 refer to the refractive index and main electronic absorption frequency of the dielectric and liquid, respectively, and ν_m is

the metal frequency parameter as defined in Lipkin *et al.*⁴⁵ We used the following parameters: $a = 30$ nm; $\kappa = 1/8$ nm; $\epsilon_m = 80.4 \epsilon_0$; $\epsilon_0 = 8.85 \times 10^{-12}$ F m⁻¹; $\zeta = 1$; $\psi_s = -47$ mV and -54 mV for Au and Ag NPs, respectively; $\psi_s = -37$ mV for the PDDA-PSS layer;⁴⁶ $n_1 = 1.45$; $n_3 = 1.33$; $\nu_1 = 3.2 \times 10^{-15}$ s⁻¹; $\nu_3 = 3 \times 10^{-15}$ s⁻¹; $\nu_m = 6.2 \times 10^{-15}$ s⁻¹ for Au and 6.2×10^{-15} s⁻¹ for Ag.⁴⁵

Confocal Detection of NPs. It should be noted that the detected signal is interferometric.³³ Depending on the scanning wavelength, the NPs can appear as bright or dark spots in the images. For localization purposes, the scanning wavelength was always set in resonance (*i.e.*, 405 nm for Ag NPs and 532 nm for Au NPs), regardless the wavelength used for printing the NP. In this way, the NPs are detected as bright spots in all cases.

Calculation of Printing Error. The radial square mean deviation was calculated as

$$S_r = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 + (y_i - \bar{y})^2}$$

where $\bar{x}$ and $\bar{y}$ are the average positions in the x and y directions, respectively.

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Notes

The authors declare no competing financial interest.

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